RESEARCH SYNTHESIS FOR PROVEIT

Quadratic batching, index absorption, and smaller consistency equations

Explicit Diophantine forms whose unsolvability is equivalent in ZFC to Con(ZFC)

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Abstract

We give three compatible improvements over the Diophantine consistency forms at the pinned ProveIt repository revision. First, we test its 22 quadratic residuals simultaneously, using a 32-element Sidon set, instead of first forming a quartic sum of squares and then using exponential positional weights. The Jones interface yields a polynomial in nine nonnegative integers of exact degree 166 177 600 and an 868-operation evaluator; the strictly positive version uses 864 operations. A companion through the existing Sun interface has ten unrestricted integer unknowns, exact degree 449 607 016, and 773 operations. Second, a short-range Pell-index argument absorbs an index quotient into the existing norm carrier, giving eleven positive unknowns at degree 1154, or a two-equation system of degrees 2 and 1152; a further positive translation gives degree 1140. Third, complementing whole tile labels replaces bitwise clauses and reduces the fixed native coefficient payload by about 33%.

All constants have finite recipes. The native tuple and source size constants also have materialized binary payloads. The archive contains explicit division-free circuits, a complete proof of the new reductions, dependency records, and independent finite certificates. The mathematical equivalences use the inherited native compiler and literal K215 calibration. The ten-integer companion also uses Sun's 2026 preprint. This report does not claim an end-to-end new proof-assistant theorem, global minimality, or a proof of consistency.

Source revision: `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`

Repository: `https://github.com/VladimirReshetnikov/ProveIt`

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1 Results, comparison, and logical meaning

There is no single useful ordering of "simplest" Diophantine forms. The number of unknowns, degree, number of equations, arithmetic circuit size, and size of fixed coefficients are distinct resources. We therefore give explicit forms on several parts of that tradeoff, with a common fully specified coefficient tuple.

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$. All additional letters introduced in formulas are polynomial abbreviations, not existentially quantified unknowns. A circuit operation means one binary addition, subtraction, or multiplication. Squaring and multiplication by a fixed coefficient count; shared registers and fixed leaves do not. Powers in displayed formulas always have fixed nonnegative integer exponents and are expanded into multiplication gates in the delivered circuits.

Theorem 1.1 (Explicit consistency forms)**.** *With the four fixed native ports specified in Section 5, the archive defines polynomials F_9, F_9^+, F_{10}, and F_{11} such that ordinary mathematics formalizable in ZFC proves*

$$\begin{aligned} \mathrm{Con(ZFC)} &\longleftrightarrow \neg\exists \mathbf{x} \in \mathbb{N}^9 \; F_9(\mathbf{x}) = 0, \\ &\longleftrightarrow \neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^9 \; F_9^+(\mathbf{x}) = 0, \\ &\longleftrightarrow \neg\exists \mathbf{x} \in \mathbb{Z}^{10} \; F_{10}(\mathbf{x}) = 0, \\ &\longleftrightarrow \neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{11} \; F_{11}(\mathbf{x}) = 0. \end{aligned}$$

The last line also has an equivalent two-equation form in the same eleven unknowns. The first two lines use the parameterized Jones ratio theorems. The third additionally uses the stated Lucas and interpolation results of Sun [4]. Every line uses the inherited ordinary compiler and literal K215 calibration at the pinned revision [1].

Form	Domain	Unknowns	Exact degree(s)	Operations
Quadratic batching	$\mathbb{N}$	9	166 177 600	868
Positive translation	$\mathbb{Z}_{>0}$	9	166 177 600	864
Sun companion	$\mathbb{Z}$	10	449 607 016	773
Index absorption	$\mathbb{Z}_{>0}$	11	1154	123
Two-equation form	$\mathbb{Z}_{>0}$	11	2, 1152	119
Shifted two-equation form	$\mathbb{Z}_{>0}$	11	2, 1152	118
Translated product	$\mathbb{Z}_{>0}$	11	1140	129
Translated shifted system	$\mathbb{Z}_{>0}$	11	2, 1138	124

Table 1: New forms in the delivered archive. The certificate circuit for the nine-variable degree proof is intentionally a different evaluator.

1.1 What improves over the pinned repository

The nine-variable degree previously reached

$$92324 \cdot 5^{32} + 296 = 2\,149\,585\,634\,469\,985\,961\,914\,062\,796.$$

The new degree is smaller by a factor of approximately $1.294 \cdot 10^{19}$. The 864-operation positive circuit also improves the separate historical 1473-operation positive circuit by 41.34%; the degree-optimized package 37 circuit uses 1500 operations. These are separate baseline comparisons, not an invented baseline combining the best features of different evaluators.

The ten-integer baseline in package 31 has degree

$$126496 \cdot 5^{31} + 27064 = 589\,042\,901\,992\,797\,851\,562\,527\,064$$

and 1442 operations. Quadratic batching reduces the degree by approximately $1.310 \cdot 10^{18}$ and the circuit cost by 46.39%. The domain remains unrestricted integers.

The eleven-positive construction removes one supplied unknown from the direct twelve-positive Pell-norm family. It trades degree and arithmetic for that reduction: the repository still has a twelve-positive degree 1022 construction with 124 operations and a separate 108-operation degree 1081 construction. The present work does not improve the historical 67-operation single equation or the 59-operation system. It also leaves intact the usefulness of the 31-positive quadratic system and its quartic sum of squares when low degree is the primary goal.

The native ports now use 339,476 raw bytes rather than 506,664, a 32.998% reduction. Their largest bit length falls from 1,016,946 to 681,374. This is a reduction in native compiler constants; it is not a claim about the size of a fully expanded 166-million-degree polynomial.

1.2 The three ideas

The first change is to preserve the system's quadratic structure. Multiplying a quadratic coefficient kernel by the square of a digit code can select a residual by one coefficient. A Sidon set gives unambiguous quadratic monomials with maximum position 799. Disjoint translated blocks select all 22 residuals simultaneously. A single binary carry test enforces every target coefficient to be zero, so there is no need to take a sum of squares before entering the few-variable interface. This reduces both the degree and the work needed to evaluate the source code. It also removes the old enormous Jones ceiling constant: the nine-variable source can use simply $b = \epsilon + 2$ and $B = \kappa b^2$.

The second change is arithmetic rather than combinatorial. The eleven-variable norm initially recovers only a weakened divisibility $E \mid H(k - R - 1)$. Cancelling H here would be wrong. Instead we recover the main Pell index first, place the remaining index in an interval shorter than $E/\gcd(E, H)$, and deduce the exact equality $2n = R + 1$. This restores the full divisibility and reconstructs the deleted positive quotient. The proof explicitly avoids a predecessor bound that required the deleted quotient. A further positive translation of the same carrier lowers degree 1154 to 1140 at the cost of six operations; it is supplied as a separate optional form.

The third change uses a whole-label complemented encoding. An occupied label x is represented by $2^\ell - 1 - x$. A selected tile contributes $x + 1$, so one guarded binary field checks the entire label by requiring its low ℓ bits to vanish. This replaces one guard field per bit by one per label while retaining the occupied high bit needed by the native compiler. We recompute the complete coefficient tuple and a compatible packed-rank controller; changing only a few constants would not suffice.

1.3 What the consistency claim means

Here Con(ZFC) means that no finite syntactic ZFC refutation exists, using the effective proof presentation calibrated in the repository. The fixed K215 machine satisfies the inherited theorem

$$\text{K215 halts on blank input} \quad \longleftrightarrow \quad \text{a finite ZFC refutation exists.}$$

The named repository endpoint is `KEndpoint215.blank_halts_iff_zfc_refutation`. The compiler and arithmetic reductions connect a halt with an integer zero, in both directions. Negation gives Theorem 1.1.

This is a theorem about equivalence. It does not prove either side of the consistency statement. In particular it creates no conflict with the second incompleteness theorem. All witnesses are

ordinary finite integers; none is a variable exponent, an unspecified enumeration index, or an oracle for consistency.

The archive separates ordinary proofs, formal source endpoints, and finite checks. We have not rebuilt the entire upstream proof-assistant development or formalized the new batching and absorption arguments in its kernel. The checks establish the finite facts claimed in their receipts; they do not establish an unbounded equivalence by testing. There is also no claim that nine is minimal for this particular consistency predicate, or that the Sidon set or operation counts are globally optimal.

2 Direct quadratic batching and a nine-variable equation

2.1 The exact positive quadratic source

We start with the literal 31-variable source of package 30, retained by the later report packages [1]. Its strictly positive unknowns, in fixed order, are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_a, \Delta.$$

Only in this subsection, put $q = \beta J + 1$, $k = \eta + \zeta$, $H = 4a + 3$, and $C_w = q - F - \alpha$. The fixed ports are (β, K_0, M_C, M_F) from Section 5. The 22 residuals, each required to vanish, are

$$\begin{aligned}
&L - q^2 + 1, && N - q(L+1),\\
&X - wq, && a - E - sN,\\
&E - X(a-E), && c - k(a-E) - \eta,\\
&P - E(c-\eta) - k, && \tau^2 - P(P-k) - 1,\\
&D - X - ac - \gamma H, && \Delta - a^2 - H,\\
&C_2 - c^2, && D^2 - \Delta C_2 - 1,\\
&k - jc + V - 1 - hE, &&
\end{aligned} \tag{2.1}$$

$$\begin{aligned}
&jc - V - (q-F)(N-q) + C_w L - (M_C + qM_F)J,\\
&(K_0 + w)C_w + q - F - z(q-1) - 1,\\
&F_2 - f^2, \qquad V - of + c, \qquad g - y^2 + V^2,\\
&I - iC_2, \qquad B_a - I^2, \qquad B_a - \Delta(F_2 - 1),\\
&y^2 - B_a g - 1.
\end{aligned}$$

Let $r_0, \dots, r_{21} \in \mathbb{Z}[n_1, \dots, n_{31}]$ be these residuals after replacing every positive source coordinate by $n_i + 1$. The delivered literal parent file `nine/parent31.json` has SHA-256

`1c336ba0b3c9827ffddcf7ce895cad9dca2907f907cde7484c7f387bb2cb1287.`

Its historical binding metadata is preserved as provenance; the active binding of this report is the new complete sibling compiler tuple.

For clarity, the elimination producing the 31-variable source is exact on the positive domain. Its former register was $G_p = (q-1)(q-F) + \alpha = q(q-F) - C_w$. On $N = q(L+1)$ one has

$$LG_p = (q-F)(N-q) - C_w L.$$

The transport row gives $(K_0 + w)C_w + (q-F) = 1 + z(q-1) \geq q$. If $C_w \leq 0$, its left side is at most $q - F < q$, a contradiction. Thus $C_w > 0$, $q - F > 0$, and the reconstructed G_p is positive.

The unchanged normalized compiler theorem therefore gives

$$\exists \mathbf{n} \in \mathbb{N}^{31} \bigwedge_{j=0}^{21} r_j(\mathbf{n}) = 0 \quad \longleftrightarrow \quad \text{K215 halts on blank input.} \tag{2.2}$$

The fifth residual at $\mathbf{n} = 0$ is 1, so a simultaneous zero is never the all-zero vector. We use this small fact in completeness.

2.2 A finite Sidon certificate replaces exponential positions

Use the following increasing list of 32 positions:

$$\begin{aligned}(0, a_1, \dots, a_{31}) = (&0, 10, 35, 55, 62, 105, 144, 145, 159, 168, 232, 245,\\ &306, 323, 325, 353, 371, 374, 382, 494, 499, 525, 566, 578,\\ &610, 647, 681, 685, 741, 777, 783, 799).\end{aligned} \tag{2.3}$$

All 528 unordered pair sums with repetition are distinct. The file `weights.json` and the independent verifier give a complete finite certificate. No unverified extremality claim for this set is needed. Put

$$M = 799, \qquad d = 4M + 1 = 3197, \qquad \mathcal{H} = 22d = 70334.$$

Write

$$r_j(\mathbf{n}) = \sum_{|\alpha| \le 2} p_{j,\alpha} \mathbf{n}^\alpha, \qquad d_{j,\alpha} = (2 - |\alpha|)! \prod_{i=1}^{31} \alpha_i!\, p_{j,\alpha}.$$

Define the following fixed polynomials and fixed target positions:

$$\begin{aligned} E_j(T) &= \sum_{|\alpha| \le 2} d_{j,\alpha} T^{2M - \sum_i a_i \alpha_i},\\ E_*(T) &= \sum_{j=0}^{21} T^{jd} E_j(T), \qquad\qquad Z_*(T) = \sum_{j=0}^{21} T^{jd+2M}. \end{aligned} \tag{2.4}$$

There are 136 nonzero weighted coefficient records and only 16 distinct nonliteral coefficient recipes. They are all explicit polynomials in the four fixed native ports.

Lemma 2.1 (Simultaneous quadratic extraction)**.** *For* $c(T) = 1 + \sum_{i=1}^{31} n_i T^{a_i}$,

$$[T^{jd+2M}]\, c(T)^2 E_*(T) = 2 r_j(\mathbf{n}) \qquad (0 \le j < 22).$$

Proof. Pad a monomial of total degree at most two with zero positions to obtain a pair from the set in (2.3). The pair-sum certificate makes this representation unique. The coefficient of its monomial in $c(T)^2$ is $2!/((2 - |\alpha|)! \prod_i \alpha_i!)$; this cancels the factorial weight in $d_{j,\alpha}$, leaving $2 p_{j,\alpha} \mathbf{n}^\alpha$. Finally $c(T)^2 E_j(T)$ has support in $[0, 4M]$. Translating it by jd puts it in $[jd, jd + 4M]$; these intervals are disjoint. Thus no other row contributes to a target. □

The important distinction is between a polynomial's degree and the choice of a positional code for its monomials. A quadratic system needs unique pair sums, not separate base-five positions for a quartic in 31 coordinates. Moreover the 22 target tests can share the same unknown digit code; one does not need an extra unknown for each row.

2.3 One explicit size constant

Expand each weighted coefficient into monomials in the four positive native ports. Let C_* be one plus the sum of the absolute values of all scalar coefficients times their corresponding port monomials, over all 136 records. Thus every $|d_{j,\alpha}| < C_*$. This is a bounded, finite integer expression, independent of the existence of any witness. Set

$$\kappa = 2^{\mathrm{bitlength}(2048C_*)} > 2048C_*. \tag{2.5}$$

For the delivered tuple,

$$\mathrm{bitlength}(C_*) = 1\,359\,514, \qquad \boxed{\kappa = 2^{1\,359\,525}.}$$

The constant's unsigned little-endian binary magnitude has 169,941 bytes; its compressed payload and all recipe hashes are included. This replaces any unspecified coefficient-height parameter. The derived fixed coefficients, including negative ones, remain signed integer leaves of the final circuit.

2.4 The new source code and its two-way interpretation

For the moment let $b \geq 2$ and $g > 0$ be integers. Set

$$\begin{aligned}
B &= \kappa b^2, & Q_0 &= B^{M+1}, & \ell &= \sum_{i=1}^{31} B^{a_i},\\
\lambda &= \sum_{t=0}^{\mathcal{H}-1} B^t, & S_0 &= 2(1+g)^2 E_*(B) + B\lambda,\\
S &= g + Q_0 S_0, & T &= Q_0 - 1 - (b-1)\ell + Q_0(B-2)Z_*(B),\\
N &= B^{\mathcal{H}+M+2} = B^{71135}, & R &= S(N^2-N) + (T+1)(N^2-1),\\
G_{\mathrm{cap}} &= 32bB^M.
\end{aligned} \tag{2.6}$$

Everything in this display is an integer polynomial in b, g with fixed integer coefficients. In particular the finite sum defining λ introduces no division in the polynomial.

Lemma 2.2 (Unconditional nonvanishing and conditional size)**.** *For every $b \geq 2, g > 0$, the integer R is nonzero. If additionally $g < 4bB^M$, then $0 \leq S, T < N$ and $R \geq N^2 - 1 \geq N \geq \max\{b, 8\}$.*

Proof. Since $Q_0 \mid N$, reduction of (2.6) modulo Q_0 gives

$$R \equiv (b-1)\ell \pmod{Q_0}.$$

The positions in ℓ are distinct and at most M, while $b < B$. Consequently $0 < (b-1)\ell < Q_0$, proving $R \neq 0$ without a sign assumption on $E_*(B)$ or R.

Now suppose $g < 4bB^M$. We have $g < Q_0$, because $B > 4b$. The highest exponent in E_* is at most $\mathcal{H} - 2M - 1$. Using $B \geq 2$ and the coefficient bound,

$$|E_*(B)| < 2C_* B^{\mathcal{H}-2M-1}, \qquad 1 + g < 5bB^M.$$

Thus

$$|2(1+g)^2 E_*(B)| < 100 C_* b^2 B^{\mathcal{H}-1} < B^{\mathcal{H}}.$$

Since $B^{\mathcal{H}} \leq B\lambda < 2B^{\mathcal{H}}$, it follows that $0 < S_0 < 3B^{\mathcal{H}}$. The low mask $Q_0 - 1 - (b-1)\ell$ is between 0 and $Q_0 - 1$: its base-B digits are $B - b$ at the chosen positions and $B - 1$ elsewhere. Also $(B-2)Z_*(B) < B^{\mathcal{H}}$. The extra factor of B in N therefore gives $0 \leq S, T < N$. Finally $T + 1 \geq 1$ and $S \geq 0$ imply $R \geq N^2 - 1$; the remaining bounds are immediate. □

Write $\tau_2(u,v)=0$ for addition without a binary carry, equivalently disjoint sets of occupied binary positions for nonnegative u,v.

Lemma 2.3 (All residuals in one carry test)**.** *Suppose b is a power of two, $b\ge 2$, and $0<g<4bB^M$. Then*

$$\tau_2(S,T)=0 \quad\longleftrightarrow\quad \begin{cases} g=\sum_i n_i B^{a_i}, \quad 0\le n_i<b,\\ r_0(\mathbf{n})=\cdots=r_{21}(\mathbf{n})=0. \end{cases}$$

Conversely, every simultaneous natural zero can be encoded in this way for arbitrarily large powers of two b, with $0<g<bB^M$.

Proof. Now B and Q_0 are powers of two. The strict block bounds in Lemma 2.2 split the carry condition into

$$\tau_2\big(g, Q_0-1-(b-1)\ell\big)=0, \qquad \tau_2\big(S_0,(B-2)Z_*(B)\big)=0.$$

The first mask has all bits set at unchosen digit positions. At a chosen position its digit $B-b$ has precisely the bits above the lowest $\log_2 b$ bits set. Hence it gives exactly the stated code and digit bounds.

For decoded digits let h_t be the coefficient of T^t in $c(T)^2E_*(T)$. Its degree is at most $\mathcal{H}-1$, and

$$|h_t|\le C_*\left(1+\sum_i n_i\right)^2<1024C_*b^2<B/2.$$

Therefore the integer

$$S_0/2=(1+g)^2E_*(B)+(B/2)\lambda$$

has ordinary base-B digits $B/2+h_t$, all strictly between 0 and B. Both summands in the second carry test are even. Divide them by two; the mask becomes $(B/2-1)Z_*(B)$. At each chosen target digit, $B/2-1$ occupies all the lower binary positions. A digit strictly between 0 and B is disjoint from this mask exactly when it equals $B/2$. By Lemma 2.1, the correction at the target is $2r_j(\mathbf{n})$, so every residual vanishes.

For completeness choose a power of two b exceeding all coordinates of a given zero, and use its digit code. Since the zero is not the zero vector, $g>0$. The geometric estimate using $B>b$ gives $g\le(b-1)\sum_{e=1}^{M}B^e<bB^M$. The same digit calculations, now in the forward direction, prove both carry tests. The choice of b can be made arbitrarily large. □

The central-binomial carry lemma of Jones [2], provided as `Jones1982.lemma_2_16` in the pinned `Carries.lean`, states that for $N=2^a$ and $0\le S,T<N$,

$$\tau_2(S,T)=0 \quad\longleftrightarrow\quad N^2\mid\binom{2R}{R}, \qquad R=S(N^2-N)+(T+1)(N^2-1). \tag{2.7}$$

This theorem uses precisely the source expression (2.6). It needs no quartic index or base-five normalization. The source construction is now complete.

2.5 The full nine-variable polynomial

The nine nonnegative unknowns are

$$(\epsilon,g_0,h_0,s_0,w_0,\phi,i_0,j_0,n)\in\mathbb{N}^9.$$

Put $b = \epsilon + 2$ and $g = g_0 + 1, h = h_0 + 1, s = s_0 + 1, w = w_0 + 1, i = i_0 + 1, j = j_0 + 1$. Use R, N, G_{cap} from (2.6) and define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & P &= 2(RY)^2 U,\\
K &= R + 1 + h(P-1), & A &= RY(U+1), & B_1 &= 2R+1,\\
W_1 &= bw, & C &= B_1 + W_1 + \phi, & J &= 4A - 5,\\
Q &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2-1)C^2 + 1, & &\\
E &= iJDC^2, & F &= (A^2-1)E^2 + 1, & &\\
G_p &= A + F(CD + 1 - A), & H &= B_1 + jC, & I &= (G_p^2 - 1)H^2 + 1,\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI, & u &= JF,\\
v &= QF + J(H - C), & & & &\\
m &= K^2 G_{\mathrm{cap}} - 4(C - KY)^2 G_{\mathrm{cap}} - 8gK^2. & & & &
\end{aligned} \tag{2.8}$$

Here m is a register; the ninth unknown is n. Next set

$$\begin{aligned}
\sigma &= Y(U+1), & r &= W_1 + \phi,\\
M_D &= AC + \sigma(r+1)^2 + 1, & M_F &= AE + 1,\\
T_{CH} &= CH + r^2, & M_I &= F(DT_{CH} + 2H^2 + A^2) - A^2 + 1,\\
V_0 &= PK + 1, & V_1 &= M_D M_F M_I + 1,\\
M_{\mathrm{div}} &= Q^2 + (H-C)^2 + 1, & \Omega &= u(2m-1),\\
\Theta &= u(n + M_{\mathrm{div}}) + v - \Omega(2M_{\mathrm{div}} + V_0 V_1). & &
\end{aligned} \tag{2.9}$$

The required equation is

$$\boxed{F_9 = \left(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\right)^2 - 4\Omega^4 V_0^2 a_0 a_1 = 0.} \tag{2.10}$$

Equations (2.3)–(2.10), together with the four native ports, specify a literal integer-coefficient polynomial. The JSON arithmetic circuit is its expanded straight-line definition. No square roots, tests, divisions, or variable exponents occur in it.

2.6 Signed bounds before positive source recovery

It is essential not to assume $R > 0$ when applying the final polynomial. For a freely chosen large g, $E_*(B)$ can make R negative. The following proof makes the order of implications explicit.

By Lemma 2.2, $R \neq 0$ for every assignment of the eight outer natural coordinates. Since $U, Y \geq 1$,

$$|A| \geq 2, \quad P \geq 2R^2 \geq 2, \quad K \geq R + P \geq 2R^2 + R \geq 1.$$

Consequently $J \neq 0$, $AJ > 0$, $D \geq 1$, $AE \geq 0$, and $F \geq 1$. The completed-square bounds from package 37 apply with no sign restriction on R. For completeness, here are their short proofs. Because $A = R\sigma$ and $C = 2R + r + 1$,

$$\begin{aligned}
M_D - AC &= \sigma(r+1)^2 + 1 > 0,\\
M_D + AC &= \sigma\big(4R^2 + 2R(r+1) + (r+1)^2\big) + 1 > 0.
\end{aligned}$$

The quadratic expression is nonnegative; twice it is $(2R + r + 1)^2 + 4R^2 + (r+1)^2$. Thus $M_D \geq |AC| + 1$ and $D = (AC)^2 - C^2 + 1 \leq M_D^2$. Similarly $M_F = AE + 1 \geq 1$ and $F \leq M_F^2$. Since $H = (j+1)C - r$,

$$T_{CH} - CH = r^2 \geq 0, \qquad T_{CH} + CH = C^2 + (C-r)^2 + 2jC^2 \geq 0.$$

It follows that $T_{CH} \geq |CH|$. Using $G_p = FDC + F - A(F-1)$, the integer inequalities $|H| \leq H^2$ and $|AH| \leq A^2 + H^2$ imply

$$|G_p H| + |H| + 1 \leq FDT_{CH} + 2FH^2 + (F-1)A^2 + 1 = M_I.$$

Thus $M_I \geq 1$ and $\sqrt{|I|} \leq M_I$, even when the candidate I is negative. Altogether

$$V_0, V_1 \geq 2, \qquad \sqrt{|a_t|} \leq V_t - 1 \quad (t = 0, 1). \tag{2.11}$$

Indeed $1 \leq a_0 \leq (PK)^2$ and $\sqrt{|a_1|} \leq M_D M_F M_I$. Also $u \neq 0$ and

$$\left|\frac{v}{u}\right| = \left|\frac{Q}{J} + \frac{H-C}{F}\right| \leq |Q| + |H-C| \leq M_{\mathrm{div}} - 1. \tag{2.12}$$

All these statements precede square conditions, divisibility, and $m > 0$.

Lemma 2.4 (Signed-divisor finalizer)**.** *Let integers $a_0, a_1, u, v, m, M, V_0, V_1$ satisfy $u \neq 0$, $M \geq 1$, $|v/u| \leq M - 1$, $V_0, V_1 \geq 2$, and $\sqrt{|a_t|} \leq V_t - 1$. Define Ω, Θ, and F_9 as above, with $M_{\mathrm{div}} = M$. Then*

$$\exists n \in \mathbb{N}\ F_9 = 0 \quad \longleftrightarrow \quad a_0, a_1 \text{ are integer squares}, \quad u \mid v, \quad m > 0.$$

Proof. The odd integer $2m - 1$ is nonzero, so $\Omega \neq 0$. Choose complex roots $r_t^2 = a_t$. Direct multiplication factors F_9 into

$$\prod_{\varepsilon,\delta \in \{-1,1\}} (\Theta - \Omega(\varepsilon r_0 + \delta V_0 r_1)).$$

A zero gives one vanishing factor. Absorb its signs in the chosen roots, and put $s = \Theta/\Omega = r_0 + V_0 r_1 \in \mathbb{Q}$. If $s \neq 0$, the identity $2sr_0 = s^2 + a_0 - V_0^2 a_1$ makes both roots rational. A rational number with integer square is an integer. If $s = 0$, the bounds give $|r_1| = |r_0|/V_0 < 1$, hence the integer $|a_1| < 1$; then $a_1 = a_0 = 0$. This includes negative candidate radicands and coincident square classes, without presuming independent field automorphisms.

Set $Z = 2M + V_0 V_1 + r_0 + V_0 r_1$. The root bounds give $Z \geq 2M + 1 > 0$. Dividing the vanishing factor by the possibly negative u gives

$$n + M + v/u = (2m - 1)Z.$$

Its left side is at least $n + 1 > 0$, so $m > 0$. Its right side is integral, so $u \mid v$. Conversely, if these three conditions hold, choose integer roots and define $n = (2m-1)Z - M - v/u$. This is an integer at least 2, and makes the corresponding factor vanish. □

2.7 Soundness and completeness of the nine-variable form

The parameterized ratio theorems in `NineRatio.lean` have the following precise interface. For $R, N \geq 8$, $0 < b \leq \min\{R, N\}$, and $G > 0$, the conditions

$$a_0, a_1 \text{ squares}, \qquad JF \mid QF + J(H - C), \qquad m > 0$$

with $h, s, w, i, j > 0$ and $\phi \geq 0$ imply

$$N^2 \mid \binom{2R}{R}, \qquad b \text{ is a power of two}, \qquad 8g < G. \tag{2.13}$$

Conversely, the first two conditions in (2.13) together with $32g < G$ produce positive h, s, w, i, j and nonnegative ϕ satisfying the three displayed ratio conditions. These are

`Jones1982.Nine.ratio_sound` and `ratio_complete`; the formulas match (2.8) with $G = G_{\mathrm{cap}}$. They quantify arbitrary admissible R, N, b, G and do not require Jones's original source packing.

Suppose $F_9 = 0$. The unconditional bounds and Lemma 2.4 first give $m > 0$. As $K \geq 1$ and the subtracted square term in m is nonnegative, this already implies

$$8g < G_{\mathrm{cap}}, \qquad g < 4bB^M.$$

Only now apply Lemma 2.2: it supplies positive R, N and every size hypothesis of the ratio theorem. Apply (2.13), then (2.7), then Lemma 2.3. This gives a zero of the 22 parent residuals, and hence a blank K215 halt by (2.2).

Conversely suppose the parent residuals have a natural zero. Choose b and its positive code $g < bB^M$ by Lemma 2.3. Its carry condition and (2.7) give the binomial divisibility; all ratio size conditions hold. Since $32g < 32bB^M = G_{\mathrm{cap}}$, `ratio_complete` supplies the six remaining outer witnesses. Lemma 2.4 supplies $n \in \mathbb{N}$. Translating the positive outer witnesses down by one gives a zero of F_9. We have proved

$$\exists \mathbb{N}^9 \; F_9 = 0 \quad \longleftrightarrow \quad \exists \mathbb{N}^{31} \; \bigwedge_j r_j = 0,$$

and thus the first line of Theorem 1.1.

For strictly positive coordinates use

$$(\epsilon_+, g, h, s, w, \phi_+, i, j, n_+) \in \mathbb{Z}^9_{>0}, \quad b = \epsilon_+ + 1, \quad \phi = \phi_+ - 1, \quad n = n_+ - 1.$$

This coordinate bijection gives F_9^+, preserves exact degree, and saves four operations in the delivered evaluator. It does not extend the result to nine unrestricted integers.

2.8 Exact degree, with cancellations resolved

Treat fixed coefficients as degree zero. The actual degree cannot be read from the original nested expression by blindly taking maxima: its highest apparent terms cancel. The following exact identities expose the surviving leading term. Put $t_0 = \sigma(r+1)^2$ and

$$\begin{aligned}
\delta_D &= M_D^2 - D = 2ACt_0 + t_0^2 + 2AC + 2t_0 + C^2, \\
\delta_F &= M_F^2 - F = 2AE + E^2, \\
W_- &= M_I - G_p H \\
&= FDr^2 + 2FH^2 + (F-1)A^2 + 1 - FH + A(F-1)H, \\
\delta_I &= M_I^2 - I = W_-(M_I + G_p H) + H^2 - 1, \\
\delta_V &= V_1^2 - DFI \\
&= \delta_D M_F^2 M_I^2 + D\delta_F M_I^2 + DF\delta_I + 2M_D M_F M_I + 1.
\end{aligned}$$

With $E_0 = u(n + M_{\mathrm{div}}) + v - 2\Omega M_{\mathrm{div}}$, the inner square in (2.10) is exactly

$$E_0^2 - 2E_0\Omega V_0 V_1 + \Omega^2 V_0^2 \delta_V - \Omega^2 a_0. \tag{2.14}$$

Some decisive total degrees are as follows:

N	142270	R	426808
P	1707239	K	1707240
A	995890	C	426808
D	2845396	F	11381586
I	30161198	a_1	44388180
V_0	3414479	V_1	22194090
Ω	16362637	δ_D	1991784
δ_F	9389806	δ_I	29307585
δ_V	43534568	E_0	17216257

The first three terms defining δ_V have degrees 43 534 568, 42 396 400, and 43 534 567. The first uniquely dominates. The third term in (2.14) uniquely dominates, at degree 83 088 800. The final subtracted term of (2.10) has strictly smaller degree than twice this value. Hence

$$\boxed{\deg F_9 = 166\,177\,600.} \tag{2.15}$$

For a nonvanishing certificate, substitute all nine natural inputs by t. The coefficient of $t^{166177600}$ is exactly

$$\boxed{2^{196}\kappa^{83088652} > 0.}$$

The verifier tracks at every gate a total-degree upper bound and the exact coefficient of the corresponding diagonal power. All sixteen derived native coefficients remain independent symbolic leaves; none is assigned zero or presumed generic. The final coefficient is independent of them. This proves the claimed degree for the actual materialized tuple as well as for every compatible specialization.

The production circuits use sparse Horner evaluation of E_*, shared repeated squaring, and a divide-and-conquer geometric-sum identity. Their literal gate counts are 868 for $\mathbb{N}^9$ and 864 for $\mathbb{Z}^9_{>0}$. The separate cancellation certificate uses 917 operations; it is a proof artifact, not the claimed fastest evaluator.

3 A companion equation in ten unrestricted integers

The same quadratic coefficient packing also compresses the ten-integer construction already present in source 31 of the repository [1]. The resulting polynomial has

$$\boxed{\text{10 integer unknowns,} \qquad \deg F_{10} = 449\,607\,016, \qquad \text{773 binary operations.}}$$

The earlier source 31 equation has degree 589 042 901 992 797 851 562 527 064 and 1,442 operations. The comparison uses the same convention: every binary addition, subtraction and multiplication is charged, including multiplication by a fixed coefficient; repeated registers and fixed leaves are free. Thus this application improves both degree and circuit size, while retaining the same ten unrestricted integer unknowns. It adds the Lucas-theoretic dependency on Sun's 2026 preprint [4]; the nine-natural-variable construction does not need that preprint.

3.1 A different source scale, with the same quadratic kernel

Use the same 31 shifted quadratic coordinates and 22 residuals r_j, the same Sidon positions $0, a_1, \dots, a_{31}$, and the same $M = 799$, $d = 3197$, $H = 70334$ as in the preceding packing construction. With $|\alpha| \le 2$, write

$$r_j(\mathbf{n}) = \sum_{\alpha} p_{j,\alpha}\mathbf{n}^{\alpha}, \qquad c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, p_{j,\alpha}.$$

Define the fixed polynomial and target mask

$$E(T) = \sum_{j=0}^{21} \sum_{|\alpha|\le 2} c_{j,\alpha} T^{jd+2M-\sum_i \alpha_i a_i},$$

$$Z(T) = \sum_{j=0}^{21} T^{jd+2M}.$$

There are 136 nonzero weighted coefficient records and 16 distinct nonliteral coefficient recipes. All four native ports are bound to the new tuple compiled from complemented labels in the sibling `compiler` directory.

For each monomial $q\prod_i \mathrm{port}_i^{e_i}$ in these records, majorize its magnitude by $|q|2^{\sum_i e_i b_i}$, where b_i is the actual bit length of the corresponding native port. Let C_{S} be one plus the sum of all these majorants. This bit-length majorant is distinct from the exact positive-monomial majorant used for the nine-variable source. The delivered exact arithmetic gives

$$\mathrm{bitlength}(C_{\mathrm{S}}) = 1\,359\,515.$$

Set

$$\beta_{\mathrm{S}} = 2^{1\,359\,528} > 2048C_{\mathrm{S}}.$$

Its exponent is even, so $\beta_{\mathrm{S}} \equiv 1 \pmod 3$. The binary payload of this additional constant occupies 169,942 bytes before compression and 200 bytes in the delivered compressed form. The number is fixed by the finite compiler data, independently of whether a refutation exists.

For integer variables f, g, define

$$b = 1 + 3f, \qquad \mathcal{B} = \beta_{\mathrm{S}} b^2, \qquad \ell = \sum_{i=1}^{31} \mathcal{B}^{a_i},$$

$$\lambda = \sum_{t=0}^{H-1} \mathcal{B}^t, \qquad N_0 = \mathcal{B}^{M+1}, \qquad N_1 = 4\mathcal{B}^H,$$

$$N = N_0 N_1 = 4\mathcal{B}^{71134}, \qquad \mathscr{C} = b\mathcal{B}^M, \qquad M_0 = N_0 - 1 - (b-1)\ell.$$

Put

$$J_0 = 2(1+g)^2 E(\mathcal{B}) + \mathcal{B}\lambda,$$

and then

$$S_0 = g + N_0 J_0, \qquad T_0 = M_0 + N_0(\mathcal{B}-2)Z(\mathcal{B}),$$

$$R_0 = (S_0 + T_0 + 1)N + T_0 + 1, \qquad X = (N-1)R_0, \qquad Y = N^2.$$

Here E is a polynomial abbreviation rather than a new unknown. All powers and summation limits are fixed. No sign restriction is imposed on these polynomials for arbitrary free inputs: the reverse bridge first recovers the bounds used in the next paragraph.

The source contract. These formulas have the following three properties. If the quadratic system has a natural solution, there are arbitrarily large $f > 0$ and a g with $b \le g < \mathscr{C}$, such that b is a square and a power of two, and $Y \mid \binom{2X}{X}$. Conversely, $f \neq 0$, $0 \le g < 2\mathscr{C}$, the same two conditions on b, and that divisibility imply a natural solution of the system. Finally, whenever $f \neq 0$, b is a square and $0 \le g < 2\mathscr{C}$, one has $X > 3b$ and $Y > \max\{b, 256\}$, even before imposing the power-of-two condition.

Proof of the changed source step. A square $b = 1 + 3f$ with $f \neq 0$ has $b \ge 4$ and $f > 0$. The highest exponent of E is at most $H - 2M - 1$, so

$$|E(\mathcal{B})| < 2C_{\mathrm{S}}\mathcal{B}^{H-2M-1}.$$

For $0 \le g < 2b\mathcal{B}^M$, the inequality $1 + g \le 3b\mathcal{B}^M$ gives

$$|2(1+g)^2 E(\mathcal{B})| < 36C_{\mathrm{S}} b^2 \mathcal{B}^{H-1} < \mathcal{B}^H.$$

Since $\mathcal{B}^H \le \mathcal{B}\lambda < 2\mathcal{B}^H$, we obtain $0 < J_0 < 3\mathcal{B}^H < N_1$ and $g < N_0$. The mask M_0 has digits $\mathcal{B} - b$ at the positions a_i and $\mathcal{B} - 1$ elsewhere, so $0 \le M_0 < N_0$. Also $(\mathcal{B}-2)Z(\mathcal{B}) < N_1$. Consequently $0 \le S_0, T_0 < N$, $R_0 \ge N + 1$, $X \ge N^2 - 1 > 3b$, and $Y > \max\{b, 256\}$. Moreover $N \equiv 1 \pmod 3$, so $3 \mid X$.

If b is a power of two, so are $\mathcal{B}, N_0, N_1, N$. Sun's central-binomial carry lemma [3, Lemma 2.2] gives

$$Y \mid \binom{2X}{X} \quad \Longleftrightarrow \quad \tau_2(S_0, T_0) = 0.$$

The two blocks split this into the code-validity test $\tau_2(g, M_0) = 0$ and the residual test $\tau_2(J_0, (\mathcal{B}-2)Z(\mathcal{B})) = 0$. The first test decodes $g = \sum_i n_i \mathcal{B}^{a_i}$ with $0 \le n_i < b$ by [3, Lemma 2.4], which explicitly permits arbitrary distinct digit positions. For such digits, put

$$K_0 = (1+g)^2 E(\mathcal{B}) + (\mathcal{B}/2)\lambda, \qquad J_0 = 2K_0.$$

Every coefficient of K_0 lies strictly between 0 and $\mathcal{B}$, because its signed correction has magnitude at most

$$C_{\mathrm{S}} \left(1 + \sum_i n_i\right)^2 < 1024 C_{\mathrm{S}} b^2 < \mathcal{B}/2.$$

The selected digits are $\mathcal{B}/2 + 2r_j(\mathbf{n})$. Adding $(\mathcal{B}/2-1)Z(\mathcal{B})$ without a binary carry forces each selected digit to equal $\mathcal{B}/2$, hence every $r_j(\mathbf{n}) = 0$. This proves soundness. Conversely, a simultaneous zero is nonzero because the fifth residual at $\mathbf{n} = 0$ equals 1. Choose $b = 4^q$ larger than all its coordinates, set $f = (b-1)/3$, and encode it in g. Then $b \le g < \mathscr{C}$, and both carry tests hold. This proves completeness, with q arbitrarily large. No sum of squares is used in this source step.

3.2 The precise imported Lucas interface

The positive and reverse bridge proofs in [4, Propositions 6.1 and 6.2] use precisely the three-part source contract of its Proposition 2.1, established above for the new source, the fixed form $b = 1 + 3f$, and the monomial bound $\mathscr{C} = 2^{(1\,359\,528)M} b^{2M+1}$. Their proofs do not otherwise use the old base-five coefficient code. We use this proof-level specialization, rather than asserting that the new source is literally Sun's original source polynomial. The positive bridge supplies a Lucas certificate, one square and one divisibility condition; the reverse bridge recovers the source contract from those conditions and a positivity inequality. Sun's modified Lucas certificate [4, Theorem 3.1] and full residue interpolation [4, Theorem 4.1] are additional imported results. The latter permits the second Lucas witness to lie in any prescribed residue class modulo a suitable first witness. This permits a positivity witness and the divisibility merger to share variables. The positive bridge is the specialization $p = P = 2$, $Q = 1$ of [3, Theorem 4.12]; the reverse bridge uses the estimates of [3, Theorem 4.13], replacing its Lucas-certificate call by [4, Theorem 3.1].

The complete ten-variable list is

$$(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

Use the source quantities above and define

$$\begin{aligned} L &= lY, & U &= 2LX, \\ V &= 4gwY, & W &= bw, \\ A &= U(V+1), & B_1 &= 2X+1, \\ C_1 &= B_1 + (A-2)h, & P &= U^2 V, \\ K &= X + 1 + k(P-2), & S &= 2A-5, \\ T &= 3WC_1 - 2(W^2-1), & r_1 &= (P^2-4)K^2 + 4. \end{aligned}$$

Next set

$$\begin{aligned} \Delta &= 8\mathscr{C}^3 g K^2 - g^2\left(32(C_1 - KL)^2 \mathscr{C}^3 + g^2 K^2\right), \\ O &= f^2 l^2 \Delta, \qquad \xi = zx. \end{aligned}$$

The modified Lucas quantities are

$$\begin{aligned} D &= (A^2-4)C_1^2+4, & E_1 &= C_1^2 D\xi, \\ F &= 4(A^2-4)E_1^2+1, & G &= 2(A+2)(A-2)^2E_1^2-1, \\ \Omega_1 &= 16E_1, & H_1 &= C_1+(B_1-C_1)F+\Omega_1 Fy, \\ I &= (G^2-1)H_1^2+1, & r_2 &= DFI, \\ r_0 &= b\big(1+(3O-4)b^2z^2\big). \end{aligned}$$

The positive bridge has $O \geq 4$, since $g \geq b \geq 4$ and its positive integer Δ is divisible by g. As proved in source 31 [1], the coprimality of b and $1+(3O-4)b^2z^2$, together with $b \equiv 1 \pmod 3$, makes r_0 square and $z \neq 0$ enforce b square and $O > 0$. Conversely, a Pell solution with its second coordinate divisible by b supplies such z.

3.3 The final polynomial and its implication in both directions

Let

$$\begin{aligned} \pi(v) &= (v^3-v-1)(v^2+23), & \Sigma &= \xi S^5, \\ t &= T+Sy, & \Theta &= (t^3-tS^2-S^3)(t^2+23S^2). \end{aligned}$$

Define the weights and scaled radicands by

$$w_0 = 1, \qquad w_1 = r_0, \qquad w_2 = r_0(P^2+K^2+3),$$

$$Z_0 = \Sigma m - \Theta, \qquad A_i = \Sigma^2 w_i^2 r_i \quad (0 \leq i \leq 2),$$

and put

$$V_0 = Z_0^2 + A_2 - A_0 - A_1, \qquad W_0 = V_0^2 + 4Z_0^2A_2 - 4A_0A_1.$$

The requested integer equation is

$$\boxed{F_{10} = W_0^2 - 16V_0^2Z_0^2A_2 = 0.}$$

Every symbol other than the ten listed inputs is a polynomial abbreviation or a fixed integer.

The local quintic merger and norm arguments are inherited from source 31 [1]. Byte-exact copies of that proof, its mathematical review and its finalizer are supplied under `signed10/inherited/`; the historical coefficient binding there is inactive. For clarity, their exact roles are as follows. The polynomial π has no rational root and has a root modulo every positive integer. Thus $\Theta \neq 0$, since $S = 2A-5 \neq 0$. Divisibility $\Sigma \mid \Theta$ forces $\xi \neq 0$ and $S \mid T$: if $v_p(S) > v_p(T)$, the two homogeneous factors have valuations $3v_p(T)$ and $2v_p(T)$, contradicting divisibility by S^5. Conversely, if $S \mid T$ and $\xi \neq 0$, a suitable residue class of y modulo $|\xi|$ makes the divisibility true. Full residue interpolation selects that class while preserving the Lucas square certificate. This converse is existential in y, not a pointwise assertion for every y.

The norm identity is

$$F_{10} = \prod_{\varepsilon \in \{\pm 1\}^3} \left(Z_0 - \Sigma \sum_{i=0}^{2} \varepsilon_i w_i \sqrt{r_i} \right).$$

At a zero, $\Sigma = 0$ would imply $F_{10} = \Theta^8 \neq 0$, so $z \neq 0$. Then $|r_0| > 1$, and the structural weights satisfy

$$|w_1| > \sqrt{|r_0|}, \qquad |w_2| > \sqrt{|r_0|} + |w_1|\sqrt{|r_1|}.$$

A rational signed sum of these roots must have each root rational: a Galois automorphism changing the largest indexed root cannot cancel its change with the smaller terms. Rational

square roots of integers are integers, so the three radicands are squares and $\Sigma \mid \Theta$. This reasoning also covers negative candidate radicands, zero roots, and repeated square classes.

The reverse Lucas bridge therefore gives a simultaneous natural zero of the 22 residuals. Conversely, its positive bridge, the coprime Pell step and full residue interpolation give the three square conditions and $\Sigma \mid \Theta$; an integer m then kills one norm factor. Together with the native compiler and calibrated K215 endpoint, this proves

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}^{10}\ F_{10}(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

This is an equivalence based on the stated mathematical dependencies, not a proof of consistency or a new proof-assistant certificate.

3.4 Degree and actual-instance checks

The degrees of X, Y are 426,804 and 284,536. The three scaled radicands A_0, A_1, A_2 have degrees 13,947,118, 24,764,324 and 112,401,754. The last dominates the squared norm center and the other radicands, so the final leading part is A_2^4. The claimed exact degree is therefore $4 \cdot 112\,401\,754 = 449\,607\,016$. Independently, diagonal leading coefficients for the actual complemented tuple are nonzero modulo 101, 1009 and 1,000,003: they are 81, 287 and 270,876. These are exact finite certificates that the structural degree bound is attained after specializing all fixed coefficients.

The verifier independently reconstructs the 136 records by evaluating quadratic values, gradients and homogeneous parts at the all-one coordinate vector, while the builder uses binomial expansion. It checks all 528 pair sums, every native-port hash and strict coefficient bound, the materialized β_{S}, the formal norm expansion, and 1,620 named-register comparisons against literal scalar formulas over three finite fields. These checks establish the finite artifact's transcription and quantitative claims; the unbounded equivalence rests on the proofs above and the stated imported theorems.

4 Eleven positive unknowns: absorb the index quotient

The second new mechanism removes an actual supplied variable from the direct Pell construction. It gives a single equation in eleven positive integers, of exact degree 1154 and cost 123 binary operations, or two equations of maximum degree 1152 and cost 118 after the inherited transport shift. The predecessor direct family uses twelve positive integers and three equations. This is a different tradeoff from the nine-variable Jones construction: its advantage is a moderate degree and a small explicit circuit. The comparison is with the pinned repository collection [1], not a global minimality assertion.

The old index quotient h is reconstructed from a divisibility condition inside the already quantified carrier. The first divisibility recovered is weaker than the original one. A short-range argument for the Pell index makes it sufficient. This is why simply deleting the old index equation would not have worked.

4.1 The polynomial and its exact supplied domain

Use the entire fixed coefficient tuple (β, K_0, M_C, M_F) of the compiler, with M_F the shifted high mask. The eleven supplied unknowns are exactly

$$J, F, \alpha, z, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.$$

All other letters below are polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & s_0 &= s + 3q, & C_w &= q - F - \alpha,\\
X &= wq, & Y &= s_0 q^3, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H, & \ell &= EkY,\\
A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2, & b_0 &= X + ac,\\
I &= ic^2, & D &= 1 + \Delta I^2, & Q &= \Delta(D-1) = (\Delta I)^2,\\
d &= Q - 1, & v &= cT, & b &= c + RD.
\end{aligned}$$

Here R and the transport expression are

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_t &= (K_0 + w)C_w + q - F - z(q-1).
\end{aligned}$$

Define the sharp weight and new carrier by

$$\begin{aligned}
e &= Q(Dv^2 + b^2) - 1, & B_a &= 2Qvb,\\
C_* &= E + 2(X + Y) + s_0 q^2, & M &= I(2T + RC_* ic),\\
L_0 &= 1 + HM, & S &= HE,\\
L &= EL_0, & W &= Lb_0 + H(Et + k - R - 1).
\end{aligned}$$

The finite conjugate product

$$\mathcal{N} = \prod_{\substack{g=\pm\sqrt{G},\ \tau=\pm\sqrt{A}\\ f=\pm\sqrt{D}}} \Big[d(W - Lg - SM\tau)^2 - S^2 e + S^2 B_a f \Big] \tag{4.1}$$

is an integer polynomial. Its eight formal factors define a polynomial identity; the displayed radicals are not additional unknowns, and independent formal signs are not assumed to be independent field automorphisms. The single equation and two-equation form are respectively

$$\boxed{N_t(1 + \mathcal{N}) - 1 = 0}, \qquad \boxed{N_t = 1, \quad \mathcal{N} = 0}. \tag{4.2}$$

For a literal division-free evaluation, put

$$\begin{aligned}
n &= (SM)^2 A, & d_n &= dn,\\
P_H &= S^2 QDv^2, & U_H &= S^2 Qb^2,\\
e_H &= P_H + U_H - S^2, & j &= 4P_H U_H,\\
\mathcal{T} &= 2d(W^2 + L^2 G), & \kappa &= [d(W^2 - L^2 G)]^2,\\
\mathcal{A} &= 2(d_n + e_H), & \mathcal{B} &= (d_n - e_H)^2 - j,\\
a_r &= \mathcal{T} - \mathcal{A}, & b_r &= \mathcal{B} - \kappa,\\
v_r &= 16 d_n j - a_r(a_r \mathcal{T} + 2b_r), & u_r &= b_r^2 - a_r^2 \kappa,\\
\mathcal{N} &= \kappa v_r^2 - u_r(\mathcal{T} v_r - u_r).
\end{aligned}$$

The system compares the two final terms directly. The identity follows by reducing a quartic modulo the quadratic with root sum $\mathcal{T}$ and root product κ, exactly as in the predecessor norm; it remains valid when quadratic fields coincide or a radicand specializes to a square.

4.2 Recover roots without assuming the deleted equation

At a root of the product equation, N_t and $1+\mathcal{N}$ are integer units. For $x_0 = d(W^2 + L^2G)$, one has

$$\mathcal{T} = 2x_0, \qquad \kappa = x_0^2 - 4d^2W^2L^2G, \qquad \mathcal{N} \equiv (u_r - x_0v_r)^2 \pmod 4.$$

Thus $1+\mathcal{N} \neq -1$, and $\mathcal{N} = 0$, $N_t = 1$. The authentic transport interface now gives

$$q \geq 16, \quad C_w > 0, \quad 3q < R < q^4 - 2, \quad Y > 3q^4 > 3R, \quad X \geq 16, \quad E > R + 1. \tag{4.3}$$

Also $k \geq 2$ and $kY < c < k(Y+1)$. For example, $C_w \leq 0$ would imply $N_t \leq 1-F-(z-1)(q-1) \leq 0$. We do *not* assume $k > E$ or $R < k$: those predecessor bounds depended on the omitted equation.

Lemma 4.1 (Uniform auxiliary bound)**.** *Every coherent complex auxiliary branch satisfying*

$$f^2 = D, \qquad dy^2 = e - B_af = Q(vf - b)^2 - 1$$

obeys $|y| < M\sqrt{A}$ *under* (4.3).

Proof. Since $\Delta > R + 2$ and $I \geq c^2$,

$$Q - b = \Delta I^2(\Delta - R) - c - R > 0, \qquad \sqrt{D} < \Delta I < d.$$

For $Z = v\sqrt{D} + b > 1$,

$$|y|^2 \leq \frac{QZ^2 + 1}{d} < [Z(1 + 1/d)]^2,$$

the last numerator difference being $QZ^2 - d > 0$. Consequently

$$|y| < v\sqrt{D} + b + v + 1 < cT((a + 2)I + 1) + c + RD + 1. \tag{4.4}$$

Set $\rho = (1 + 1/Y)/E$. Then $c/\ell < \rho$ and $(a + 3)\rho < 2$. The exact identity

$$(a + 2)^2\rho = E + 2Y + X + 6 + \frac{Y}{X} + \frac{5}{X} + \frac{4}{Y} + \frac{8}{E} + \frac{4}{EY}$$

has its last four terms less than 1. As $Y/X = s_0q^2/w \leq s_0q^2$ and $X \geq 16$, it is less than $C_* - 2$. Divide (4.4) by $\sqrt{A} > \ell$, and use $(c + 1)/\ell < 2$:

$$\frac{|y|}{\sqrt{A}} < 2TI + R(C_* - 2)i^2c^3 + 2 \leq M.$$

No lower bound on k beyond $k \geq 2$ was needed. □

The same outer bounds give $G > A$ and $b_0 > \sqrt{A}$: use $\Delta c^2 > k^2Y^2(E + Y)^2 > A - 1$ and $b_0 = \ell + E\eta + Yc + X > \ell + k > \sqrt{A}$. Every root of the inner polynomial has the form $x = M\tau + y$, hence $|x| < 2M\sqrt{A}$. If a vanishing factor in (4.1) had $g = -\sqrt{G}$, then

$$x = \frac{W + L\sqrt{G}}{S} = t + \frac{k - R - 1}{E} + (M + 1/H)(b_0 + \sqrt{G}) > 2M\sqrt{A},$$

because the first two terms have positive sum by $E > R + 1$. This is impossible. If G were nonsquare, actual quadratic conjugation would force this forbidden branch along with the positive branch. Thus g is a positive integer and $x = (W - Lg)/S$ is rational.

In a normal splitting field, an automorphism negating an irrational τ would give $2M\sqrt{A} \leq |y| + |y'| < 2M\sqrt{A}$. Thus τ is rational, hence integral. Then $y = x - M\tau$ is rational, and $f = (e - dy^2)/B_a$ is rational, hence integral. Finally the inherited rational Pell saturation lemma

gives $y \in \mathbb{Z}$: if $m \geq 2$, $V \in \mathbb{Z}$, and $m^2V^2 - (m^2-1)y^2 = 1$ with $y \in \mathbb{Q}$, then $y \in \mathbb{Z}$. It follows by clearing a reduced denominator, classifying the two resulting integer Pell solutions, and using abscissa divisibility to force ordinate divisibility. Here apply it with $m = \Delta I$ and $V = vf - b$. The complete argument, including zero cases, is reproduced in the accompanying proof file. In particular x is integral. The carrier equality implies

$$E \mid H(k - R - 1). \tag{4.5}$$

At this stage it would be invalid to cancel H.

4.3 Recover the main index before the index quotient

Let $A_0 = a + 2$, and write χ_{A_0}, ψ_{A_0} for the usual Pell sequences. The fundamental positive integer-coordinate solution for $A_0^2 - 1$ is $(A_0, 1)$. Main and normalized strong norm recovery gives

$$g = \chi_{A_0}(p), \quad c = \psi_{A_0}(p), \quad |f| = \chi_{A_0}(m), \quad I = \psi_{A_0}(m), \qquad p, m \geq 1.$$

The standard strong-divisibility and composition identities now give

$$p \mid m, \qquad c \mid m, \qquad m \geq pc > 2p. \tag{4.6}$$

Here is the direct argument, to make its independence from the deleted equation explicit. Since $c^2 \mid I$, strong divisibility first gives $p \mid m$. Write $m = pj$. Then $\psi_{A_0}(pj) = c\psi_{\chi_{A_0}(p)}(j)$, while modulo c the second factor is $j\chi_{A_0}(p)^{j-1}$. Since $\chi_{A_0}(p)^2 \equiv 1 \pmod{c}$, we get $c \mid j$. No ordinary-strong large-c rank lemma is necessary. Also $c > 2Y > 6R$, and $p \geq 2$ with $c \geq (2A_0 - 1)^{p-1} > 2p$.

The inherited signed auxiliary-index argument can therefore be applied before the main congruence modulo H. For clarity, its essential steps are included. Put $S_0 = \Delta I$ and $V = c(Tf - 1) - Rf^2$. The auxiliary Pell equation shows that

$$S_0|V| = \chi_{S_0}(j_0), \qquad |y| = \psi_{S_0}(j_0)$$

for a positive odd $j_0 = 2r + 1$. The usual odd-index polynomial Q_r has

$$\chi_{S_0}(2r+1) = S_0 Q_r(S_0^2), \quad Q_r(1 - A_0^2) = (-1)^r \psi_{A_0}(2r+1), \quad Q_r(0) = (-1)^r(2r+1).$$

Because $Q = \Delta(f^2 - 1) \equiv 1 - A_0^2 \pmod{|f|}$ and $V \equiv -c \pmod{|f|}$, these give $\psi_{A_0}(j_0) \equiv \pm c \pmod{|f|}$. Squaring and doubling yields

$$\chi_{A_0}(2j_0) \equiv \chi_{A_0}(2p) \pmod{\chi_{A_0}(m)}.$$

Signed index reduction, using $0 < 2p < m$, gives $j_0 \equiv \pm p \pmod{m}$, hence modulo c. This follows by reducing modulo $2m$, reflecting indices to $[0, m]$, and using strict growth and $2\chi_{A_0}(m-1) < \chi_{A_0}(m)$ to exclude a negative representative. Meanwhile $Q \equiv 0 \pmod{c^4}$ and $V \equiv -R \pmod{c}$ give $j_0 \equiv \pm R \pmod{c}$. Since $p, R \in (0, c/2)$, one obtains

$$p = R. \tag{4.7}$$

4.4 A short index range restores the full divisibility

The first norm has fundamental unit $(2EY + 1, 2)$, so for $P_0 = 2EY + 1$,

$$|\tau| = \chi_{P_0}(n), \qquad k = 2\psi_{P_0}(n), \qquad n \geq 1.$$

Because $P_0 > A_0$ and $c > k$, we have $n < p = R$. Put $g_0 = \gcd(E, H)$ and $\mu = E/g_0$. Since $H = 4XY + 4Y + 3$,

$$\gcd(Y, H) = \gcd(Y, 3) \leq 3, \qquad g_0 \mid X \gcd(Y, H), \qquad \mu \geq Y/3 > R. \tag{4.8}$$

Equation (4.5) gives $k \equiv R+1 \pmod{\mu}$. As $P_0 \equiv 1 \pmod{E}$, the Pell recurrence also gives $k \equiv 2n \pmod{E}$. Hence $2n \equiv R+1 \pmod{\mu}$. But $1 \le n \le R-1$ gives $|2n-R-1| < R < \mu$. Therefore

$$2n = R+1, \qquad E \mid (k-R-1).$$

Set $h = (k-R-1)/E$. It is integral and positive: $R > 3q \ge 48$ gives $n \ge 25$ and $k = 2\psi_{P_0}(n) > 2n = R+1$.

Divide the carrier identity by E to get

$$L_0 b_0 + H(t+h) = L_0 g + H(M\tau + y).$$

Since $L_0 \equiv 1 \pmod{H}$, we obtain $g \equiv b_0 \pmod{H}$. Thus $\gamma = (g-b_0)/H$ is an integer. Moreover $n \ge 2$ gives $k \ge 4P_0 > X$, hence $c > X$, and $g - b_0 > (a+1)c - ac - X = c - X > 0$. So $\gamma > 0$. All original normalized compiler equations are now restored, including $k - hE - R = 1$ and the main congruence. The signed-quotient soundness theorem handles a possible negative f by changing f, T together, and extracts a blank K215 halt. The order of recovery above avoids every use of h before its reconstruction.

4.5 Completeness and exact degree

For a canonical positive compiler witness the inherited theory gives $X = 2^R$, $R > 3q$, and $Y > 2^R$. Since $q \ge 16$ and $2^{3q} > 3q^4$, its old coordinate $s_{\text{old}} = Y/q^3$ satisfies $s_{\text{old}} > 3q$. Hence $s = s_{\text{old}} - 3q > 0$ is legitimate. Define

$$t = L_0\gamma + M\tau + y - h.$$

This is positive because $h < k/E$ and $\tau > \ell = EkY > k/E$. It makes a coherent factor in (4.1) vanish and preserves $N_t = 1$. This proves both directions of existence equivalence. The shifted system uses

$$(K_0 + w)C_w = F + z(q-1),$$

with $z_{\text{old}} = z + 1$. Canonical witnesses have $w = 2^R/q > q > F$ and $C_w > 0$, so $z_{\text{old}} \ge 2$. This saves one gate while preserving positivity and existence.

For the degree calculation, let $\mathcal{N}_{\text{old}}$ be the predecessor's sharp half-carrier norm, of degree 1056. In the localization at E,

$$\mathcal{N} = E^{16}\mathcal{N}_{\text{old}}\left(t + \frac{k-R-1}{E}\right), \qquad s_{\text{old}} = s + 3q. \tag{4.9}$$

Each of the eight factors gains E^2, and its degree in the old carrier is at most 16, so the denominators clear. The rational shift has total degree -2, below $\deg t = 1$, and cannot change a highest homogeneous part. Writing $q_0 = \beta J$, $k_0 = \eta + \zeta$, and $s_{00} = s + 3q_0$, the predecessor's proved leading form gives

$$\mathcal{N}_{1152} = 2^{46} i^{48} k_0^{96} w^{90} s_{00}^{192} q_0^{710} (q_0 - F)^{16} \neq 0.$$

Thus $\deg \mathcal{N} = 1056 + 16 \cdot 6 = 1152$. The transport leading form is $w(q_0 - F - \alpha) - zq_0 \neq 0$, proving exact product degree 1154 and system residual degrees $2, 1152$, uniformly for the authentic $\beta > 0$.

The delivered circuit has 75 multiplications and 48 additions or subtractions for the product. Independent checks include 90 formal norm products modulo three primes, 77 carrier-denominator identities, exact integer and rational fixtures, 500 parity fixtures, and two complete univariate polynomial calculations of degree and leading coefficient. Another 320 bound fixtures deliberately satisfy $k \le E$, testing the hypothesis removed from the predecessor argument. The four actual fixed payloads are hashed and used in 40 additional modular norm checks. These diagnostics support the transcription; the unbounded proof above and the inherited compiler calibration establish the theorem in ZFC.

4.6 An optional degree-1140 companion

A further positive translation of the same eleven supplied coordinates gives a second Pareto point. Retain all definitions in Section 4, put

$$\sigma = M(\ell + 2c - X) - b, \qquad W = Lb_0 + H\big(E(t+\sigma) + k - R - 1\big),$$

and change nothing else. The product now has exact degree 1140 and 129 binary operations; the shifted two-equation version has residual degrees $2, 1138$ and 124 operations. Thus the main 123-operation construction remains the less expensive one. The optional circuits are separate files.

Soundness of the positive translation. The unchanged parity argument gives $N_t = 1$ at any product root and hence all outer bounds. Write $C = C_*$ and $B = \ell + 2c - X > 0$. Using $X/q = w \geq 1$, direct expansion gives

$$\begin{aligned} C\ell - \Delta c &= EY\{\eta(X - 6 - 1/q) + \zeta(2X - 4)\} - 4E\eta \\ &\quad + (w-1)Y^2c - (4Y+3)c \\ &\geq EY\{\eta(X - 7 - 1/q) + \zeta(2X - 4)\} - (4Y+3)c. \end{aligned}$$

Since $X, q \geq 16$ and $Y \geq 4$, both bracket coefficients are at least $X/2$. As $c < k(Y+1)$ and $(4Y+3)(Y+1) < 8Y^2$,

$$C\ell - \Delta c > kY^2(X^2/2 - 8) \geq (k/3)X^2Y^2.$$

Also $C = XY + 2X + 2Y + Y/q < 3XY/2$. Consequently

$$CB - \Delta c > C\ell - \Delta c - CX > X^2Y(kY/3 - 3/2) > 1.$$

It follows that

$$\sigma = 2TIB + Ri^2c^3(CB - \Delta c) - c - R > 0.$$

Every new positive solution therefore gives a positive old carrier $t_{\mathrm{old}} = t + \sigma$, and the main soundness proof applies directly.

Completeness. Subtracting σ from the canonical old carrier gives exactly

$$t = \gamma + M\{(\tau - \ell) - ((a+2)c - g)\} + y + b - h.$$

The two root differences obey

$$\tau - \ell = \frac{\ell k + 1}{\tau + \ell} > \frac{k}{3}, \qquad (a+2)c - g = \frac{c^2 - 1}{(a+2)c + g} < \frac{c}{a+2} < \frac{2k}{X+1} < \frac{k}{3}.$$

Here $\ell > k$, $\tau < \ell + k$, and $c < k(Y+1)$ are enough. Since $b > h$, the new t is positive and completeness follows.

Exact degree. In the predecessor's sixteen-linear-factor expansion the prefactor d^8H^{16} has degree 464, and the linear factors have degrees four times 35, twice 36, and ten times 38. The translation has degree 35. For exactly two factors, namely the choices $g = \sqrt{G}$, $\tau = \sqrt{A}$, and $y = -(b - vf) + O(\lambda^{-8})$, the translated factor becomes

$$t + M\{\ell - \sqrt{A} + (a+2)c - \sqrt{G}\} - \gamma - vf + O(\lambda^{-8}).$$

Because $\sqrt{A}-\ell = k/2+O(\lambda^{-9})$, while $(a+2)c-\sqrt{G}$ has degree -1, its leading term is $-M_0k_0/2$ of degree 28. The opposite two degree-35 factors retain their degree with twice their original leading term; the other factors are unchanged in highest degree. The translated predecessor norm therefore has degree

$$464 + 2 \cdot 28 + 2 \cdot 35 + 2 \cdot 36 + 10 \cdot 38 = 1042.$$

The factor E^{16} from index absorption adds 96; the residual rational carrier correction has degree -2 and affects none of these leaders. Thus the new norm has degree 1138 and leading form

$$2^{46} i^{48} k_0^{96} w^{90} s_{00}^{188} q_0^{700} (q_0 - F)^{16}.$$

The degree-2 transport factor yields degree 1140 for the single equation. Two complete univariate polynomial checks over different finite fields agree with this form, and 60 independent formal conjugate-product checks verify the optional circuit's norm transcription.

5 Complemented whole-label clauses reduce the fixed coefficients

The improvements in the number of unknowns can use the previous coefficient tuple. A separate, compatible reduction makes the fixed native compiler smaller. We keep the ordered 943-tile language, native positions, transport geometry, and shared-anchor synchronization of package 36 at the pinned repository commit [1]. We replace its bitwise label clauses by whole-label clauses and recompute the complete tuple and rank controller. The resulting base is

$$\rho = 2^{77}, \qquad L = 8807, \qquad d = 77L = 678139, \qquad B = \rho^L.$$

The four native ports occupy 339,476 raw bytes, versus 506,664 in package 36. The largest port has 681,374 bits, versus 1,016,946. Both reductions are approximately 33.0%. These figures measure the native ports, not the coefficients of an expanded final polynomial.

5.1 One guard bit per entire label

Lemma 5.1 (Complemented label field)**.** *Let $0 \le x < 2^{\ell-1}$ and let v be an ℓ-bit copied word. Give a selected tile coefficient $x+1$, its copied bits their ordinary binary weights, and the combined field $\ell+1$ bits. Testing that the lowest ℓ bits vanish forces the copied word to be*

$$\mathrm{code}(x) = 2^\ell - 1 - x$$

when one tile is selected, and forces it to be zero when none is selected.

Proof. With no tile the field is $v \in [0, 2^\ell - 1]$. With one tile it is $x + 1 + v$, strictly between 0 and $2^{\ell+1}$. Its only possible multiple of 2^ℓ is therefore 2^ℓ. The code is injective and its highest bit is one. The extra bit prevents overflow into the next field. □

The old clauses assigned a separate radix-four field to every copied bit. For the four label widths $9, 10, 9, 10$, this required 76 payload bits. The new fields require $10 + 11 + 10 + 11 = 42$ bits. The three two-bit anchor fields remain, and are placed first to reduce the carry bound.

The selector is decoded before this lemma is used. At most 943 tile bits can be selected, while every other clause coefficient is divisible by 1024. Thus the lowest ten bits are the exact number n of selected tiles. Disjointness with the mask 1022 forces $n \in \{0, 1\}$. This reasoning does not assume carry-free label fields before selection.

5.2 A complete finite recipe

The tile slots are west, north, east, south. Their original labels fit below 2^{ℓ_s-1}. Use the following native offsets and clause offsets:

s	W	N	E	S
o_s	1023	1013	1003	983
ℓ_s	9	10	9	10
p_s	16	47	37	26

The three native anchor positions are $1095, 2195, 3296$, with clause offsets $10, 12, 14$. Define the unchanged support

$$\mathcal{E} = \{0, \dots, 942\} \cup \{o_s + j : 0 \le j < \ell_s\} \cup \{1063, 1064, 1095, 2195, 3296\}, \qquad |\mathcal{E}| = 986.$$

For tile i with original labels $x_{i,s}$, put

$$\begin{aligned} c_i &= 1 + \sum_s (x_{i,s}+1)2^{p_s} + 2^{10} + 2^{12} + 2^{14}, \\ c_{o_s+j} &= 2^{p_s+j}, \qquad (c_{1095}, c_{2195}, c_{3296}) = (2^{10}, 2^{12}, 2^{14}), \\ c_{1063} &= c_{1064} = 0, \\ \mu &= 1022 + 2^{10} + 2^{12} + 2^{14} + \sum_s (2^{\ell_s} - 1)2^{p_s}. \end{aligned}$$

The clause mask has 57 bits and population 50. All of the following powers and sums define fixed integers:

$$\begin{aligned} Z_0 &= (2^{75}-1) \sum_{e=8792}^{8802} \rho^e + (2^{39}-1)\rho^{8803}, \\ D_C &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \sum_{e \in \mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}), \\ D_R &= \rho^{42}, \\ M_C &= B - 1 - \sum_{e \in \mathcal{E}\setminus\{1\}} \rho^e - 2\rho^{1063}, \\ M_{F,0} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} \rho^{1013+j} + \sum_{j=0}^{8} \rho^{1065+j} + \rho^{2195} + \rho^{3296} + Z_0. \end{aligned}$$

The four ports, always used together, are

$$\boxed{\beta = B - 1, \quad K_0 = D_C + BD_R, \quad M_C, \quad M_F = M_{F,0} + B - 1.}$$

In particular M_F is the shifted source mask. The unshifted mask $M_{F,0}$ is not an interchangeable port.

5.3 Why the native compiler still applies

The scalar native-basis identities are unchanged: at targets $1013 + j$, $0 \le j < 10$, the current and temporal contributions are from $983 + j$ and $1013 + j$; at $1065 + j$, $0 \le j < 9$, the current and spatial contributions are from $1003 + j$ and $1023 + j$; each tested anchor $2195, 3296$ reads the current source 1095 and its own temporal bit. At each clause target $T \in \{4393, 5495\}$, the coefficient of ρ^{T-e} in D_C is precisely c_e. The finite auditor checks all 66,062 identities, including the absence of every other contribution.

The occupied high bits and injective codes are retained, so the previous matching and occupation-propagation proof applies. The clean-band and anchor intersections remain

$$((4393 - \mathcal{E}) \bmod L) \cap ((5495 - \mathcal{E}) \bmod L) = \varnothing,$$
$$((2195 - \mathcal{E}) \bmod L) \cap ((3296 - \mathcal{E}) \bmod L) = \{0\}.$$

Indeed, 1102 is absent from $\mathcal{E} - \mathcal{E}$, whereas the difference 1101 occurs only between the two tested anchors. The inequality $L > 2 \max \mathcal{E}$ excludes new circular cases. Start first forces the source anchor, then both tested anchors; the old binary and whole-radix synchronization arguments recover alignment before global occupancy is used.

For the inherited conservative pre-decoding bound, put

$$\mathcal{M} = 988 \left(2 \sum_{e \in \mathcal{E}} c_e + 6 \right).$$

Exact integer arithmetic verifies

$$\max\{4\mathcal{M} + 8,\ 2\mu + 4,\ 2(2^{75} - 1) + 4,\ 16\} < 2^{77}.$$

Thus the uniform carry bounds still hold. The largest unshifted current exponent is $5495 + 3296 = 8791$, so the padded positions 8792 through 8803 are identically zero on intended aligned cells. The padding contributes $11 \cdot 75 + 39 = 864$ tests and the original mask contributes 122. Consequently

$$\mathrm{pc}(M_{F,0}) = 986, \qquad \mathrm{pc}(M_C) = d - 986, \qquad M_C \equiv 2 \pmod 4, \quad M_{F,0} \equiv 4 \pmod 8,$$

with $0 < M_C, M_{F,0} < B - 1$. These are identities of the actual materialized ports, not only symbolic population counts.

5.4 An explicit controller for the actual packed rank

Changing the radix requires a new completeness certificate. Set

$$\lambda = 56611, \qquad k = \lambda - 1 = 56610, \qquad d = 7 \cdot 11 \cdot 8807.$$

All four displayed primes are certified by trial division. With $H_0 = 1 + BD_C + B^2(D_R + 1)$, exact modular arithmetic gives

p	7	11	8807	56611
$\mathrm{ord}_p(B)$	3	10	629	6290
$(B^2 - 1) \bmod p$	3	2	3507	20663
$H_0 \bmod p$	6	1	3377	14311

Also $\mathrm{ord}_d(B) = 18870 \mid k$ and

$$B^k \bmod \lambda^2 = 1810929280 \neq 1, \qquad B^k \equiv 1 \pmod \lambda.$$

Hence $v_\lambda(B^k - 1) = 1$.

Pad a canonical accepting rectangle to independently large powers of λ in both directions, as allowed by the inherited language theorem. Write its width and height as w_c, h_c, and set $N = w_c h_c = \lambda^n$, $q = B^N$, $T = N/\lambda$, $C_0 = d\lambda$. Choose $N > d\lambda^2$ and $h_c > \lambda$. The exact change in actual packed rank caused by the safe dummy at cell j remains

$$\delta R = -\Gamma B^j, \qquad \Gamma = \rho^{1063}(q^2 - 1)[1 + q(D_C + BD_R + B^{w_c})].$$

The unit table proves $\gcd(\Gamma, dN) = 1$.

Binomial lifting and the order certificates imply that the first T powers of B^k are precisely the residues $1 + C_0 z$ modulo $C_0 T$. Every residue modulo dN is a Boolean subset sum of

$$\{B^{2+kj} : 0 \le j < T\} \cup \{B^3\}.$$

To see this, normalize a target by B^{-2} and subtract either zero or B to make it nonzero modulo λ. Its least residue a modulo C_0 satisfies $1 \le a < C_0 < T$ and $\gcd(a, T) = 1$. Write the target as $a + C_0 u$ and solve

$$az_0 + \frac{a(a-1)}{2} \equiv u \pmod{T}.$$

The a distinct orbit entries with additive coordinates $z_0, z_0 + 1, \ldots, z_0 + a - 1$ have the desired sum. Include B^3 exactly when the initial subtraction was used.

The largest selectable cell is at most $N - N/\lambda - \lambda + 3$. Since $w_c < N/\lambda$, its required spatial and one-cell translates remain below N and avoid the origin. Thus the inherited no-wrap argument applies. Choosing the target $(R_0 - dw_c)\Gamma^{-1}$ tunes the actual rank to $R \equiv dw_c \pmod{dN}$, as required by the helical compiler.

The unchanged mask residues give canonical $R \equiv 3 \pmod 4$. Since d, N are odd, the canonical dyadic arithmetic scale is square; the standard choice of strong Pell index preserves the odd normalized multiplier, and the population identity preserves the odd scale in $Y = sq^3$. Thus signed adapters using these simultaneous promises remain compatible. Soundness does not restrict arbitrary solutions to prime-power rectangles; this choice is only used in completeness.

The scripts `build_complemented.py` and `audit_complemented.py` reproduce the integer ports and the finite certificates. The second script does not import the first and reconstructs all four full integers independently. The complete universal adapter argument and dependency boundary are retained in `compiler/PROOF.md`. Rebinding a source polynomial changes no variable count, degree, or gate count; any derived compression constants must nevertheless be recomputed from the newly bound source.

6 Reproduction, evidence, and remaining work

6.1 What is fully specified

The four fixed native integers are materialized in `compiler/coefficients/`; the binding manifest records their unsigned little-endian raw hashes, compressed hashes, and bit lengths. The native generator starts from a literal ordered 943-tile table and the literal K215 transitions. It does not execute an unbounded machine or search for a refutation. The new nine-variable κ and ten-integer β_S are also materialized, and every other fixed leaf is an explicit finite integer polynomial in those ports. Historical binding metadata in inherited source copies is expressly inactive.

The large final polynomials are supplied as division-free directed acyclic arithmetic circuits. This representation uniquely determines each coefficient of a conventional integer polynomial, and permits evaluation without printing all expanded monomials. In particular the 864-operation figure does not charge repeated shared expressions multiple times, and it does charge all nontrivial scalar multiplications and fixed powers after expansion. This convention is the same one used for the repository comparisons.

Path	Purpose
`report.tex`, `sections/`	Complete manuscript source.
`report.pdf`	Compiled and visually inspected report.
`compiler/`	New native ports; builders, independent auditor, literal data, unbounded adapter proof, and inherited notices.

Path	Purpose
`nine/`	Source kernel, fixed recipes, natural and positive circuits, cancellation certificate, and independent bootstrap review.
`signed10/`	Ten-integer circuit, source proof, Sun interface, finite checks, and pinned inherited finalizer.
`eleven/`	Main and optional eleven-positive circuits, complete proofs, root bounds, and independent mathematical review.
`provenance/`	Pinned endpoint source and machine identity, with a manifest explaining the inherited logical boundary.
`reproduce.py`	Runs all finite builders and auditors using only the Python standard library.
`Makefile`	Convenience targets for verification and PDF build.
`MANIFEST.sha256`	Hashes of archive contents, excluding itself.

From an extracted archive, run

```
python3 reproduce.py
make pdf
```

Python 3.10 or later is sufficient for the finite arithmetic checks; no third-party Python packages are required. The PDF build uses `pdflatex` and standard TeX packages. The finite computations need no network connection or external repository checkout. The included Lean source snapshots identify imported theorems; they are not a self-contained replacement for the entire upstream Lean development.

6.2 The executed finite certificates

The compiler has a separate auditor that does not import its builder. It reconstructs all four full integers and checks the label ranges, all 66,062 scalar native-basis identities, population balance, padding safety, and the residue orders used by the rank controller. The builder additionally checks 1,157,632 whole-label decoder cases. The unbounded controller argument is proved in the manuscript and companion proof.

The nine-variable verifier independently reconstructs all 136 weighted coefficient records using values, gradients, and quadratic parts of the parent residuals at the all-one vector. It checks all 528 pair sums and all strict coefficient bounds, validates actual native payloads and the additional source constant, and compares the literal formula, both domain versions, and the cancellation circuit in 450 finite-field evaluations using the actual fixed tuple. Its exact symbolic diagonal audit yields the nonzero leading term $2^{196}\kappa^{83088652}$. It also checks 768 signed-majorant and telescoping fixtures and an exact 72-case small packing analogue with invalid-digit controls. These latter examples detect transcription errors; the universal source proof does not rest on their coverage.

The ten-integer checker independently derives the quadratic source, verifies the full fixed tuple and the even source exponent, expands the three-root norm identity formally, and compares 1620 named registers over three finite fields. Its actual-instance diagonal leading coefficients are nonzero modulo 101, 1009, and 1,000,003; the leading-degree structure therefore survives the fixed coefficient specialization. The proof-level specialization of Sun's interface is stated separately from these algebraic tests.

The main eleven-positive checker compares 90 independently multiplied formal conjugate products, 77 carrier-denominator identities, exact integer and rational fixtures, 500 parity fixtures, and two complete univariate coefficient arrays for the degree calculation. Another 320 bound fixtures

deliberately have $k \leq E$, so they exercise the case in which the predecessor's deleted index condition cannot be used. Forty additional norm checks use the actual coefficient tuple. The optional translated construction has two further complete degree checks and 60 independent norm products. The ordinary proofs and independent review notes cover both positive-domain directions.

6.3 The dependency boundary

There are four logically different layers. First, the finite source data and scalar identities are completely reproducible inside this archive. Second, the new reductions—quadratic batching, its signed bootstrap, index absorption, positive translation, and complemented label clauses—are ordinary mathematical proofs given here and in the accompanying proof files. Third, the normalized native/Pell compiler, canonical completeness choices, and literal K215 logical calibration are inherited from the pinned repository. Fourth, the unrestricted-integer companion uses the explicitly named theorems of the external Sun papers. These sources have been identified and checked for the contracts used, but are not re-proved in a new proof-assistant kernel in this work.

The ordinary proofs use finite integer arithmetic, induction, elementary algebra, and the stated inherited arithmetic and compiler theorems. Thus the displayed equivalences are formalizable in ZFC. That assertion is different from claiming that an automated checker has already accepted the complete new theorem. The source files mark this distinction consistently.

6.4 Promising next steps

The structural gain is already achieved without optimizing the Sidon set. A smaller 32-position set, or a source-aware set that separates only the needed coefficient matches, would reduce the two large-degree companions further. Such a replacement needs its own exact collision certificate and must retain the complement positions needed for all target tests. Merely assigning the 136 existing monomials distinct positions is insufficient: products with the other monomials in the digit-code square must also be excluded.

The most direct formalization target is the general simultaneous quadratic extraction and carry decoder. It is a short independent lemma between the existing quadratic source and the already parameterized ratio theorem. The signed bootstrap should be included in that formalization, since a positive-source assumption on arbitrary outer inputs would be false. For the eleven-variable construction, the central new formal lemma is the reordered Pell-index recovery followed by the short-modulus argument; the proof should not import $k > E$ from the old twelve-variable formulation.

The present results do not settle an eight-natural-variable or nine-unrestricted-integer construction for this consistency predicate. They do provide a large and reproducible improvement at the existing few-variable bounds, a new moderate-degree eleven-positive form, and a smaller coherent coefficient compiler. Those are separate, verifiable advances over the pinned report family.

References

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